

HiCE object-of-interest branch

Implementation reference — touch/untouch detection via learned candidate-object localization. July 2026.

1. Overview and motivation

HiCE currently spots touch moments from globally pooled, hand-conditioned frame features. The extension adds untouch (release) detection and gives the model a learned ability to identify the candidate objects of interest, connecting the two event types through a shared, spatially anchored object representation — without adding new temporal losses that would require tuning.

Design constraints that shaped every decision below:

- **Incremental:** temporal encoder, prediction heads, displacement head, NMS, and evaluation stay untouched. New modules are gated by a flag so the baseline reproduces bit-exact when disabled.
- **Localization must be learned by the model** (a contribution), not delegated to an off-the-shelf detector at inference time.
- **Annotation reality:** hand bboxes, candidate object bboxes (TACO subset), object masks (HOI4D), grasp labels, touch/untouch events. Object annotations are class-agnostic, not attributed to left/right hand, and persist through the video (all candidate objects are annotated at every frame, not only during contact).
- **The dataset pipeline crops frames around hands** before the model sees them; this crop logic is not modified. All new geometry lives in target rendering.

2. Core design decisions and rationale

2.1 Heatmap head, not box detection

The object branch predicts a single-channel spatial heatmap over the 7×7 feature grid rather than regressing bounding boxes. Four reasons, in order of importance:

- **Localization is an intermediate variable, not an output.** A box is a decoded, discrete prediction requiring target assignment in training and NMS-like decoding at inference; gradients cannot flow through box decoding back from the temporal task. A heatmap stays continuous and differentiable end-to-end, so the touch/untouch loss can reshape where the model looks. Localization is learned in service of event detection.
- **Uncertainty and multiplicity are native.** At the exact touch moment the object is maximally occluded by the hand — precisely when a box head would be forced to commit to hard coordinates. A heatmap can be diffuse, multi-modal (several candidates), or near-empty, and the downstream pooling degrades gracefully. A box head fails hard exactly at the events of interest.
- **It matches the annotation.** Multiple unattributed boxes/masks per frame map naturally to per-cell coverage targets. Box regression would require instance assignment that does not exist in the data.
- **Precision beyond "roughly where" is wasted.** The consumer is a feature readout at 7×7 resolution; pixel-accurate corners add no information to that readout.

2.2 Placement: after hand cross-attention

The heatmap head reads the post-cross-attention, post-FFN 7×7 map. Because that map is already hand-conditioned, "object of interest" is disambiguated from "any object" by construction: a candidate region

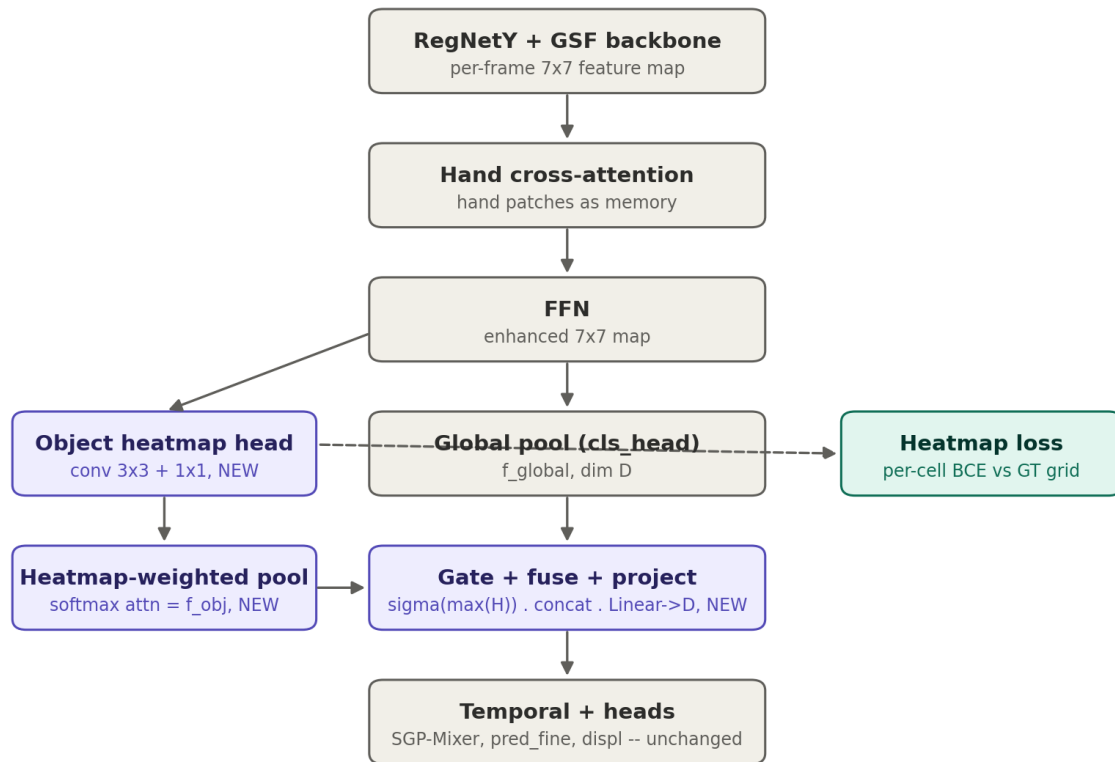
near an active hand produces different features than the same region with no hand nearby, even when the heatmap weights them equally. The heatmap answers *where* (temporally stable, since all candidates are annotated throughout); the hand-conditioned features underneath answer *when*. Do not expect or force the heatmap itself to encode event timing.

2.3 How the branch connects touch and untouch

The heatmap becomes spatial attention: softmax over the 49 cells, weighted sum of the feature map, yielding one object feature f_{obj} per frame — a learned, differentiable ROI-align. Currently, global average pooling dilutes object cells roughly 1:49 against background; the attention pool reads out features exactly where the model believes candidates sit.

Over a clip, $f_{\text{obj}}(t)$ is a trace anchored to the same spatial regions across an entire contact episode: the hand converging into the region (onset), hand-object entanglement during contact, the hand withdrawing and object appearance changing (release). The temporal encoder differentiates this trace to fire both classes, so touch and untouch are grounded in one shared quantity. The connection is delivered through the input representation with zero new temporal losses.

3. Architecture



Gray = existing HiCE (unchanged) · Purple = new modules · Teal = new loss (training only, dashed)

Forward pass. Gray modules are existing HiCE, unchanged. Purple modules are new. The dashed edge is the training-only heatmap loss.

3.1 New module specifications

- **Heatmap head:** Conv 3×3 ($D \rightarrow D/4$) → ReLU → Conv 1×1 ($D/4 \rightarrow 1$), applied to the post-FFN ($B \cdot L, D, 7, 7$) map. Output logits H of shape ($B \cdot L, 1, 7, 7$).
- **Attention pooling:** $A = \text{softmax}(H.\text{flatten}(2))$; $f_{\text{obj}} = \sum A \cdot \text{feature_map}$. Shape ($B \cdot L, D$).
- **Objectness gate:** $f_{\text{obj}} \leftarrow \sigma(\max(H)) \cdot f_{\text{obj}}$. When nothing is visible in the crop (all-zero target regime), softmax over near-uniform logits would pool background garbage; the gate smoothly shrinks the contribution to zero and the model reverts to vanilla HiCE for that frame. This is also the general robustness valve against crop hiccups.
- **Fusion:** $\text{im_feat} = \text{Linear}(\text{concat}(f_{\text{global}}, f_{\text{obj}}))$ back to dimension D , so temp_enc , the temporal stack, and all heads are untouched. Initialize so the f_{obj} half starts near zero (or a learnable scalar gate initialized at 0): at step 0 the model is HiCE and cannot regress below baseline — the debugging insurance.
- **Flag:** everything behind `--obj_head`. Off = bit-exact baseline.

4. Ground-truth heatmap rendering

4.1 Masks are not necessary

At 7×7 (32 px cells on a 224 crop) the difference between box and mask targets is below the head's resolution; a mask only tightens coverage for elongated/non-convex objects, and softmax pooling concentrates on high-coverage cells regardless. Do not put SAM-on-TACO on the critical path. If wanted later, box-prompted SAM offline gives a "box vs mask supervision" ablation row — expected gap: negligible (itself a reportable result). TACO trains from boxes; HOI4D can use masks through the same code path.

4.2 Unified recipe (boxes and masks, one code path)

- Transform annotations into cropped-frame coordinates using the same crop parameters the dataset already computes for that sample. This happens inside `__getitem__` *after* the crop is decided — targets cannot be precomputed into labels.pkl because the crop is per-sample.
- Rasterize at intermediate resolution 56×56: boxes → filled rectangles into a binary union map (overlaps handled, no double counting); masks → nearest-neighbor resize of the binary mask.
- Average-pool 8×8 → 7×7 float target in [0,1]; each cell = fraction covered by any candidate object.

```
def render_obj_target(boxes_xyxy, crop_xyxy, mask=None, out=7, inter=8):
    """boxes in original frame coords; crop_xyxy = (x0,y0,x1,y1) of the hand-based crop."""
    R = out * inter # 56
    occ = np.zeros((R, R), dtype=np.float32)
    x0, y0, x1, y1 = crop_xyxy
    sx, sy = R / (x1 - x0), R / (y1 - y0)
    if mask is not None: # HOI4D path
        m = mask[int(y0):int(y1), int(x0):int(x1)]
        occ = cv2.resize(m.astype(np.float32), (R, R), cv2.INTER_NEAREST)
    else: # TACO path
        for bx0, by0, bx1, by1 in boxes_xyxy:
            gx0 = np.clip((bx0 - x0) * sx, 0, R); gx1 = np.clip((bx1 - x0) * sx, 0, R)
            gy0 = np.clip((by0 - y0) * sy, 0, R); gy1 = np.clip((by1 - y0) * sy, 0, R)
            occ[int(gy0):int(np.ceil(gy1)), int(gx0):int(np.ceil(gx1))] = 1.0
    return occ.reshape(out, inter, out, inter).mean(axis=(1, 3)) # (7,7) in [0,1]
```

4.3 Target-side details

- **Soft targets, per-cell BCE.** Targets are in [0,1]; the loss is per-cell BCE with soft labels — the existing BCELoss semantics applied per cell (or the FocalLoss variant).
- **Foreground weighting from day one.** Typically only ~5–15% of cells are positive; without `fg_weight` (or focal), the head learns all-zeros in Stage 1.
- **Objects fully outside the crop → all-zero target.** Keep it as supervision ("nothing here" is informative); the objectness gate handles the pooling side.
- **Geometry to track: crop only.** Training augmentations are color-only (jitter/blur); horizontal flip lives in the inference-TTA path where the heatmap is model output and no target is involved. No flip handling needed for targets during training.

5. Losses and training schedule

Total loss = existing (main event loss + displacement + optional grasp) + λ_{obj} · masked per-cell BCE on the heatmap, with $\lambda_{\text{obj}} \approx 0.1\text{--}0.2$ (same weighting convention as grasp_loss). Per-frame validity mask zeroes the term on unannotated frames (largely unnecessary given dense annotation, but keep the mechanism).

5.1 Two-stage schedule (default)

- **Stage 1 — localization only.** Train backbone + hand cross-attention + heatmap head with only the heatmap loss (temporal loss weight 0). Per-frame objective: fast convergence, large batches fine. Debug localization in isolation: visualize heatmaps on ~20 clips before proceeding.
- **Stage 2 — full objective.** Load Stage 1 weights, enable fusion, train the full HiCE objective plus the heatmap loss at small weight to keep localization alive. Reuse the existing pretrain / update_pred_head loading pattern.
- **End-to-end variant** = skip Stage 1. Keep as a flag; it is a free ablation row.

6. Code diff surface

File	Change	~Lines
dataset/frame.py	Load object boxes/masks per frame; render_obj_target inside __getitem__ after crop params are known; add obj_target (+ validity mask) to the batch dict. Mirrors the existing labelID plumbing.	~40
model/model.py	_obj_head conv stack, softmax attention pool, objectness gate, fusion Linear(2D → D); gated by --obj_head; add obj_heatmap to the res dict.	~35
epoch() in model.py	One masked per-cell BCE term added to loss; one entry in loss_dict; λ_{obj} arg.	~10
main.py / args	--obj_head, --obj_loss_weight, --obj_stage flags.	~5
Everything else	Temporal stack, pred_fine, displacement, NMS, eval_from_json — untouched.	0

7. Validation checklist and risks

7.1 Before any training (order matters)

- **Visual unit test of target rendering.** Overlay rendered 7×7 targets (upsampled) on the actual cropped frames for ~10 clips from *each* dataset. One hour of work; catches every coordinate bug this pipeline can produce. Do this before the model sees a single gradient.
- **TACO coordinate conventions.** Object boxes come from a different annotation pass than the hand crops — verify resolution assumptions and xyxy vs xywh match what the crop parameters expect.
- **Object size vs grid resolution.** Compute the GT box area distribution. If the median box covers < 1 cell at 7×7 (small utensils, buttons), tap the stage-3 RegNetY features (14×14) for the heatmap head only — decide before Stage 1, not after.

7.2 During training

- Track a localization metric alongside mAP: pointing-game accuracy (heatmap argmax inside any GT region) or per-cell AP. It isolates whether a bad result is a localization failure or a fusion failure — critical when debugging time is short.
- Stage 1 heatmaps should look sensible before Stage 2 starts; if they are all-zeros, revisit `fg_weight` first.
- With the zero-initialized fusion gate, Stage 2 step-0 metrics must match the HiCE baseline. If they do not, the fusion wiring is wrong.

7.3 Known risks

- 7×7 too coarse for small objects → mitigation: 14×14 tap (checked in 7.1).
- Hand-based crop excludes the object in some frames → handled by all-zero targets + objectness gate; no crop-logic changes.
- Fusion destabilizes the temporal task → fallback: aux-loss-only variant (heatmap loss without fusion) still shapes features via deep supervision, with literally zero inference-path change.

8. Ablation plan / paper story

- Baseline HiCE (flag off, bit-exact).
- + heatmap loss only, no fusion (deep-supervision effect).
- + fusion (full method).
- Two-stage vs end-to-end training.
- Box vs mask supervision (HOI4D both available; TACO boxes, masks optional via box-prompted SAM).
- Report localization quality (pointing game / per-cell AP) and qualitative heatmap figures across an episode — approach, contact, release — anchored on the same object. These figures are the visual argument that touch and untouch are grounded in a shared object representation.